

Code-Layout, Readability and Reusability

Date	27 June 2024
Team ID	SWTID-2026-9346
Project Name	Voice Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python PEP-8 indentation standards followed.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for UI, models, utilities, reports, and assets.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names like <code>audio_file</code> , <code>transcript_text</code> , <code>semantic_score</code> .
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as <code>transcribe_audio()</code> , <code>generate_pdf()</code> , <code>analyze_audio()</code> .
5	Code Comments	Inline and block comments explain logic	Partial	Major modules are documented; minor helper functions have fewer comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Independent modules for transcription, analysis, visualization, and reporting.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Utility functions reused across the application.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Handles invalid audio files, API errors, and file upload exceptions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Audio Upload Module	Streamlit, Python	Audio input interface	High
2	Speech-to-Text Module	Whisper AI	Audio transcription	High
3	Semantic Analysis Module	Sentence-BERT	Concept similarity evaluation	High
4	Audio Feature Extraction Module	Librosa	Fluency and speech analysis	Medium
5	PDF Report Generator	ReportLab	Report generation and download	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4.5	Well organized project structure.
Readability	5	Easy to understand and maintain.
Reusability	5	Modules can be reused in future projects.
Documentation / Comments	5	Important functions are documented.
Overall Score	4.87/5	High- quality, maintainable code.